VIOLA Exposure Time Calculator (ETC)

User Manual (App User Edition)

VIOLA Team

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https://viola-etc.streamlit.app/

Scope of this document. This manual explains how to use the VIOLA ETC web application to estimate signal-to-noise ratio (SNR) performance and inspect related diagnostic outputs. It is intended for users who run the ETC through the hosted Streamlit app. Mathematical formulation, source code, supporting data files, software architecture, developer workflows, and deployment procedures are described here only when necessary. Otherwise, they are deferred to a future *Full Manual*.

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1 Overview

1.1 What VIOLA ETC is

VIOLA ETC is a Python-based modeling tool that predicts the signal-to-noise ratio (SNR) achievable for the VIOLA (VIPA-based Instrument for an Oxygen-Loaded Atmosphere) instrument under user-defined observing conditions. VIOLA is a high-throughput, ultra-high-resolution near-infrared spectrograph ($R \approx 300{,}000$; 1.1–2.5 μm, JHK bands), designed to support studies of molecular signatures in exoplanet atmospheres. The ETC computes an end-to-end photon budget from source to detector, incorporating telescope and instrument throughput, telluric atmospheric transmission, and key noise contributions including sky emission, detector noise, and thermal background. The core SNR formalism is based on, and adapted from, the exposure time calculator framework developed for IGRINS by HAN Le et al. (2015). In addition to the primary SNR outputs, VIOLA ETC includes a first-order approximation for planetary molecular signal detectability under simplified assumptions. The ETC is available as a Streamlit-based web application, providing an interactive interface for exploring instrument performance and supporting rapid iteration during development.

1.2 Typical use cases

- Estimate SNR versus wavelength for a target with a given magnitude and observing setup.
- Compare observing strategies (e.g., exposure time, number of exposures, nod subtraction settings).
- Identify dominant noise regimes (stellar photon noise vs background vs detector terms).
- Explore a simplified molecule detectability estimate using template spectra.

1.3 What the ETC does not claim

The ETC outputs are intended for forecasting and trade studies. They are not a substitute for full end-to-end simulations, injection–recovery tests, or a detection claim for exoplanet atmospheres. Any molecule-related outputs should be interpreted as a first-order estimator under explicit assumptions.

2 Access and Requirements

2.1 How to access

The VIOLA ETC is currently available only as a hosted web application:

https://viola-etc.streamlit.app/

2.2 User requirements

There are currently no special requirements beyond access to a web browser. The app is tested primarily on desktop or laptop web browsers. It is not yet validated on mobile phones or tablets. If you encounter any limitations, please let us know.

2.3 Availability notes

Because the app is hosted, performance can depend on server availability and resource limits. If the page does not load or returns an error, try refreshing the page and re-running the calculation.

3 Quick Start: 60-second workflow

1. Open the app: https://viola-etc.streamlit.app/
2. Set **Target** inputs (band, magnitude system, and target (stellar) magnitude, plus optional spectral and molecular-detection settings).
3. Set **Observing** inputs (telescope diameter, exposure time per exposure, number of exposures, and observing-condition assumptions).
4. Click **Run ETC**.
5. Check the key outputs:
    - **Results**: median SNR and SNR versus wavelength, plus a magnitude-sweep view.
    - **Sky model**: atmospheric transmission and sky background behavior.
    - **Signal & Noise**: breakdown of the signal component and the noise budget.
    - **Molecules**: first-order detectability estimate and the required nights or transits for a target detection significance.
    - **Debug**: input summary for reproducibility and issue reporting.

4 User Interface Guide

4.1 Target section

This section defines the astrophysical input signal and the wavelength window used for the SNR calculation, and it also includes optional settings for estimating planetary molecular detectability.

Band selection. Choose the observing band. The band sets the wavelength range used throughout the ETC and selects the corresponding sky-model region. The current implementation includes three near-infrared bands, with the following wavelength windows.

- **J band:** 1.10–1.30 μm
- **H band:** 1.50–1.70 μm
- **K band:** 1.90–2.50 μm

Magnitude system and magnitude. Select the magnitude system (**Vega** or **AB**) and enter the stellar magnitude in the selected band. The ETC scales the input source model to match this band magnitude.

Stellar effective temperature and spectral energy distribution. The ETC uses a stellar spectral energy distribution (SED) model, parameterized by the stellar effective temperature T_{eff}, to describe how the stellar flux varies with wavelength across the selected band. The SED choice affects the shape of the SNR spectrum, while the overall flux level remains anchored to the input band magnitude.

Select the SED model in the **Target (Advanced)** dropdown. The default option is **Blackbody**. An alternative option, **PHOENIX-NewEra**, uses stellar atmosphere spectra from Hauschildt et al. (2025). For **Blackbody**, the allowed input range for T_{eff} is 2000–40,000 K. For **PHOENIX-NewEra**, model grid is available for $T_{\text{eff}} = 2300$–12,000 K. If **PHOENIX-NewEra** is selected with T_{eff} outside this range, the ETC automatically falls back to **Blackbody**, as noted in the in-app help text accessed via the question-mark icon. At present, the PHOENIX-NewEra grid included in the app is limited to $\log g = 4.5$ and solar abundances.

Molecular detection. Enable **Molecular detection** in **Target (Advanced)** to activate a first-order, template-based estimate of planetary molecular detectability. The calculation is controlled by user inputs that define the assumed **planet-to-star line contrast**, the **line-selection thresholds** (minimum SNR per resolution element, minimum line depth, and minimum number of lines), the **detrending efficiency** (0–1), and the **required detection significance** used to estimate the number of nights or transits needed.

The molecular templates included with the app are precomputed using petitRADTRANS (Mollière et al.2019) under a fixed set of assumptions and should be interpreted as approximate, feasibility-level inputs. Template spectra are currently available for CH_4, CO, CO_2, and H_2O. Further improvements to the template generation and underlying assumptions are planned for future versions.

4.2 Observing section

This section defines the telescope aperture, integration strategy, and observing-condition inputs used to compute the photon budget and SNR.

Telescope aperture. Select the telescope diameter used for the calculation. The current web app provides two options, **2 m** and **8 m**. The telescope diameter sets the collecting area and therefore scales both the target signal and sky background. In general, larger apertures increase the achievable SNR for a fixed exposure plan, although the exact scaling depends on the dominant noise regime.

Exposure time per frame and number of frames. Set the exposure time per frame, t_{exp}, and the number of frames, N_{exp}. The total on-source integration time is

$$t_{\mathrm{tot}} = N_{\mathrm{exp}} \times t_{\mathrm{exp}}.$$

Longer total integration generally increases SNR, but the improvement depends on whether the calculation is dominated by stellar photon noise, background noise, or detector noise. In the molecular-detection mode, one transit or one night is treated as the current observing setup with the selected number of frames.

Target altitude and PWV. The **Target altitude** and **PWV** control the telluric transmission and sky background adopted by the ETC through the sky model. Lower altitude increases airmass and typically reduces atmospheric transmission. Higher PWV increases water-vapour absorption, which can strongly affect parts of the near-infrared. As a result, SNR can decrease in wavelength regions impacted by strong telluric absorption. In v0.4, the sky model is taken from a precomputed grid generated with the ESO SkyCalc Sky Model Calculator for Cerro Paranal (Noll et al. 2012; Jones et al. 2013). The grid spans PWV from 0.5 to 10 mm and target altitude from 30° to 90°. For a given user input, the app selects the closest available grid point. This sky model is intended for feasibility studies and relative comparisons, and it may not match the true observing conditions for a specific site or night. Further refinement and site-specific sky models are planned for future versions.

Seeing FWHM. The **Seeing FWHM** (arcsec) affects slit losses and therefore the fraction of stellar light that enters the spectrograph. Worse seeing relative to the slit width increases slit losses, which reduces the detected signal and can lower SNR. The current instrument defaults use a slit width of 1.6 arcsec, so seeing significantly larger than this value will typically reduce throughput.

A–B nod subtraction. If **A–B nod subtraction** is enabled, the ETC assumes observations are taken at two nod positions (A and B) and then differenced (A–B) to suppress the sky and other slowly varying background structure. Because subtraction combines the noise from both frames, the variance in the differenced image increases relative to a single exposure (approximately by a factor of two in variance, or $\sqrt{2}$ in noise). As a result, the predicted SNR can be lower than the no-subtraction case, especially when background or detector terms dominate. In practice, nod subtraction is often used to control background residuals and instrumental systematics, so the SNR trade-off should be considered together with these practical benefits.

5 Understanding Outputs

5.1 Results tab

The **Results** tab provides the primary performance outputs, including:

- The median SNR within the selected band window for the current input configuration
- SNR per resolution element as a function of wavelength across the selected band
- A magnitude-sweep view showing the estimated median SNR as a function of magnitude near the user-input magnitude under fixed observing assumptions, with optional controls for the sweep range and step size (can be enabled above the **Run ETC** button)

5.2 Sky model tab

This tab visualizes the sky-model inputs used by the ETC across the selected band. It includes the atmospheric transmission and the main background components, including zodiacal light, OH airglow, and scattered moonlight. In v0.4, the ETC uses a library of precomputed sky spectra generated with the ESO SkyCalc Sky Model Calculator for Cerro Paranal, sampled at discrete values of target altitude and PWV. For any user-selected altitude and PWV, the app adopts the nearest available grid point. As a result, the sky model is a simplified representation intended for feasibility checks and relative trade studies rather than site- and night-specific predictions. Further refinements are planned for future versions.

5.3 Signal & Noise tab

This tab visualizes the signal and the main noise terms used in the SNR calculation. The outputs include:

- Target signal in electrons, shown per resolution element and per pixel
- Background contributions, including sky emission and thermal background
- Detector-related terms, including dark current

5.4 Molecules tab (optional)

If enabled, the **Molecules** tab summarizes the template-based detectability estimate for the selected setup. For each molecule, the tab displays an overlay of the molecular template and the SNR spectrum across the band, with markers indicating the template lines used in the calculation after applying the selected thresholds. The tab reports the number of usable lines, the estimated detection significance per transit (or night, using the current number of frames by default), and the estimated number of nights or transits required to reach the user-specified detection significance. If fewer lines pass the thresholds than the **minimum number of lines**, the detection metrics are reported as `n/a`. These outputs are intended for forecasting and trade studies and should be interpreted as first-order estimates under the stated assumptions.

5.5 Debug tab

The **Debug** tab is intended to support reproducibility and issue reporting. It provides a snapshot of the run configuration, including:

- Input parameters (including values and data types)
- The last-run input set used to generate the current results

- Result metadata that can help diagnose unexpected behavior

When reporting a problem, it is helpful to include a screenshot of any error message and the relevant content from this tab.

6 Troubleshooting and FAQ

TBD